

TABLE II
SPECIFICATIONS OF SOME BUDGET-FRIENDLY OSCILLOSCOPES

Specifications	Siglent SDS1000CML+	Siglent SDS1000X	Gwinstek GDS-1000B	Gwinstek GDS-2000E	Scientech EasyScope 801C	Scientech PowerScope 824A	Keysight 2000X	R&S Scope Rider series
Bandwidth	70MHz/100MHz	100MHz	50MHz/70MHz	70MHz/100MHz	30MHz	30MHz	70MHz	60MHz
Sampling rate	1GSa/s	1GSa/s	1GSa/s	1GSa/s	Nil	Nil	2GSa/s	1.25GSa/s
Number of channels	2	2	2 or 4	2 or 4	2	2	2 or 4	2 or 4
Accuracy	DC: 5mV/div ~10V/div: ≤3.0% ≤2mV/div: ≤4.0%	DC: 5mV/div ~10V/div: ≤3.0% ≤2mV/div: ≤4.0%	DC gain accuracy: ±3%	DC gain accuracy: ±(3%) when 2mV/div or greater is selected, ±(5%) when 1mV/div is selected	±3%	±3%	±2.0% full scale	Not mentioned
Memory depth	2Mpts	14Mpts	10Mpts	10Mpts	Not mentioned	Not mentioned	1Mpts	2Mpts
Probe (active or passive)	Passive	Passive	Passive	Passive	Passive	Passive	Both	Passive
Maximum input voltage	1MQ≤400Vpk (DC+peak AC ≤10kHz), 50Q≤5Vrms	1MQ≤400Vpk (DC+peak AC ≤10kHz), 50Q≤5Vrms	300Vrms	300Vrms	400V (DC+AC peak)	400V (DC+AC peak)	Depends on scope settings; varies with different full-scale range	300Vrms
External triggering	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Display size	17.78cm (7-inch) TFT	20.32cm (8-inch) TFT	17.78cm (7-inch) TFT	20.32cm (8-inch) TFT	8cm×10cm	8cm×10cm	21.59cm (8.5-inch) TFT	17.78cm TFT
Form factor	Benchtop	Benchtop	Benchtop	Benchtop	Benchtop	Benchtop	Benchtop	Handheld
Wireless supported	No	No	No	No	Yes	Yes	No	Yes
IoT supported	Yes	Yes	LAN/Ethernet port	LAN/Ethernet port	Yes	Yes	LXI compliant with built-in Web server	Yes
Waveform update rate	Not mentioned	60,000wfms/s (normal mode) 400,000wfms/s (sequence mode)	50,000wfms/s	120,000wfms/s	Not mentioned	Not mentioned	50,000wfms/s	50,000wfms/s

budget-friendly oscilloscopes are listed in Table II.

New trends in oscilloscopes

In recent years, we have seen many changes in budget-friendly oscilloscopes. We see many innovations with added features in today's oscilloscope like better performances including higher capture rate and CPU

performance, features such as smart trigger and decoding, and mixed-signal oscilloscopes and arbitrary waveform generators.

Technological advancement in analogue-to-digital converters has resulted in high vertical resolution scopes that give better measurement accuracy.

Apart from wireless and IoT

features, the latest scopes are enhanced with logic analysers, spectrum analysers, digital voltmeters, protocol analysers, counters and so on.

In the future we could see smarter and virtual oscilloscopes with even more features at much lower cost.

To conclude, if you are planning to buy a budget-friendly oscilloscope, first check your requirement, search the Internet for options and see if you can find a value-for-money oscilloscope. But keep in mind that you should never compromise on bandwidth, signal capture rate and display. **EFY**

Major contributors to this report

- Akash Srivastava, application engineer, Keysight Technologies, India
- April Fang, sales manager, Siglent Technologies Co. Ltd, Shenzhen
- Manish Joshi, general manager - marketing, Scientech Technologies Pvt Ltd, Indore
- Srinivasa Appalla, manager - product support and application, Rohde & Schwarz India Pvt Ltd
- Sumit Sharma, marketing manager, Good Will Instrument Co. Ltd, India